

Teorija polj
Vektorski operatorji

s = skalarno polje, V = vektorsko polje

grad: $\nabla s = \left(\frac{\partial s}{\partial x}, \frac{\partial s}{\partial y}, \frac{\partial s}{\partial z}\right)$
div: $\nabla \cdot \mathbf{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$
rot: $\nabla \times \mathbf{V} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix} =$

= $(Ry - Qz, Pz - Rx, Qx - Py)$
Laplacijan: $\nabla^2 s = \frac{\partial^2 s}{\partial x^2} + \frac{\partial^2 s}{\partial y^2} + \frac{\partial^2 s}{\partial z^2}$
Smerni odvod: $\frac{\partial s}{\partial l} = grad(s) \frac{\vec{l}}{|\vec{l}|}$

Prostorska geometrija
Koordinatni sistemi

Polarni: $x = r \cos \theta, \quad y = r \sin \theta \quad J = r$
Cilindrični: $x = r \cos \theta, \quad y = r \sin \theta, \quad z = z \quad J = r$
Sferični: $x = r \sin \phi \cos \theta, \quad y = r \sin \phi \sin \theta, \quad z = r \cos \phi \quad J = r^2 \sin \phi$
 $J = \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} = \det \begin{bmatrix} x_u & x_v \\ y_u & y_v \end{bmatrix}$

Razno formule

Ločna dolžina: $s = \int_a^b |\mathbf{r}'(t)| \, dt =$
 $= \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2} \, dt$
parame. točk: $\vec{r} = r_A * t(r_B - r_A)$
Odvod integrala s parametrom:

$F(x) = \int_{u(x)}^{v(x)} f(x, y) dy \Rightarrow$
 $F'(x) = \int_{u(x)}^{v(x)} \frac{\partial(f(x, y))}{\partial(x)} dy +$
 $+ f(x, v(x)) * v'(x) - f(x, u(x)) * u'(x)$
Naravna parameterizacija:
 $|\mathbf{r}'(s)|^2 = 1, \quad s = \text{ločna dolžina}$
Tangentna premica in normalna ravnina:
 $\vec{e}$ ali $\vec{n} = \dot{\vec{r}}(t_0)$

Kot med Vektorjema: $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$

Večkratni integrali

Dvojni integral:
 $\iint_D f(x, y) \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx$
Ploščina: $A = \iint_D 1 \, dA$
Prostornina: $V = \iiint_D f(x, y) \, dA$
Težišče: $\bar{x} = \frac{1}{A} \iint_D x \, dA, \quad \bar{y} = \frac{1}{A} \iint_D y \, dA$
Masa: $m = \iiint_E \rho(x, y, z) \, dV$
Vztrajnostni moment: $I = \iiint_E r^2 \rho \, dV$

Krivuljni integrali

S. polje: $\int_C f(x, y, z) ds =$
 $= \int_{t_0}^{t_1} f(x(t), y(t), z(t)) * \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2} dt$
V. polje: $\int_C \mathbf{V} \cdot d\mathbf{r} = \int_C (P dx + Q dy + R dz)$
 $= \int_{t_0}^{t_1} (P\dot{x} + Q\dot{y} + R\dot{z}) dt$
 $d\vec{r} = \dot{\vec{r}} dt = (\dot{x} dt, \dot{y} dt, \dot{z} dt)$
Neodvisen od poti če $\vec{v}$ potencialno polje
($rot \vec{V} = 0$)

Ploskovni integrali

S. polje: $\iint_S f \, dS =$
 $= \iint_D f(\mathbf{r}(u, v)) \sqrt{EG - F^2} \, du \, dv$
kjer $E = \mathbf{r}_u \cdot \mathbf{r}_u, \quad F = \mathbf{r}_u \cdot \mathbf{r}_v, \quad G = \mathbf{r}_v \cdot \mathbf{r}_v$
 $z = z(x, y), \quad \vec{n} = (-z_x, -z_y, 1)$
V. polje: $\iint_S \mathbf{V} \cdot d\mathbf{S} =$
 $= \iint_S \mathbf{V} \cdot \vec{v} \, dS = \iint_D [\mathbf{V}, \mathbf{r}_u, \mathbf{r}_v] \, du \, dv$
 $\vec{v} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}, \quad |\vec{r}_u \times \vec{r}_v| = \vec{n}$

Integralski izreki

Greenov, Stokesov, Gaussov :
 $\oint_C (P \, dx + Q \, dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$
C - rob pozitivno orientiranega sklenjenega območja D
 $\oint_C \mathbf{V} \cdot d\mathbf{r} = \iint_S rot \mathbf{V} \cdot d\mathbf{S} = \iint_D (rot \vec{V}, \vec{r}_u, \vec{r}_v) du dv$
C je rob površine S, in D je projekcija na ravnino
 $\iint_S \vec{V} \cdot d\mathbf{S} = \iiint_W div \vec{V} \, dx \, dy \, dz$
S - zaprt površina, ki je rob območja W

Kompleksna analiza
Elementarne komp. fun.

$z = x + iy; \quad |z| = \sqrt{x^2 + y^2};$
 $\arg z = \arctan \left(\frac{y}{x} \right)$
 $e^z = e^x (\cos y + i \sin y); \quad d\varphi = \frac{dz}{iz}$
 $\log z = \log |z| + i(\arg z + 2\pi k)$
 $\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \cos z = \frac{e^{iz} + e^{-iz}}{2}$
 $\sinh z = \frac{e^z - e^{-z}}{2}, \cosh z = \frac{e^z + e^{-z}}{2}$

Integral Kompleksne funkcije

$\int_C f(z) \, dz = \int_C (u + iv)(dx + idy) =$
 $= \int_C (u \, dx - v \, dy) + i \int_C (v \, dx + u \, dy)$
 $= \int_{t_a}^{t_b} f(z(t)) \cdot \dot{z}(t) \, dt$
Cauchjeva form.: $f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz$
!Integral analitične. fun. ni odvisen od poti!

Kompleksne Vrste

Taylor: $f(x) = \sum_{n=1}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$
Geometrijska, sin, cos, e vrsta:
 $= \frac{1}{1 - z} = \sum_{n=0}^{\infty} z^n \quad ; |z| < 1$
 $\sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} \quad ; \text{za vse } z$
 $\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} \quad ; \text{za vse } z$
 $e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} \quad ; \text{za vse } z$
Laurent: $f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$
 $a_n = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz$

Analitičnost in odvedljivost in harmoničnost

C-R pogoj: $u_x = v_y \quad u_y = -v_x \quad u - x = \frac{\partial u}{\partial x}$
 $u = \int v_y \, dx = - \int v_x \, dy + C$
Pogoj harmoničnosti:
 $\nabla^2 u = \nabla^2 v = 0 / w_{xx} + w_{yy} = 0$

Formule kompleksne analize

Residum: $\oint_C f(z) dz = 2\pi i \sum \text{Res}(f, z_k)$
 $Res f(z) = \frac{1}{(m-1)!} * \lim_{z \rightarrow z_0} \left[\frac{d^{m-1}}{dz^{m-1}} ((z - z_0)^m f(z)) \right]$
[m]-pol n-te stopnje
L.V.: $f(z) = \sum_{n=-\infty}^{\infty} C_n (z - z_0)^n$
 $C_{-1} = Res f(z) \parallel f'(z)$ za stopnje ničel
Eul. form. v $\mathbb{C}$: (realna rac. funkcija [f])
 $e^{i\phi} = \cos \phi + i \sin \phi \quad z = x + iy$
 $\sin(\phi) = \frac{e^{i\phi} - e^{-i\phi}}{2i}$
 $\cos(\phi) = \frac{e^{i\phi} + e^{-i\phi}}{2}$
 $z = e^{i\phi} |d\phi = dz / iz| \int_0^{2\pi} f(\sin(\phi), \cos(\phi)) d\phi$
 $\frac{z - z^{-1}}{2i} = \sin(\phi) \mid \frac{z + z^{-1}}{2} = \cos(\phi)$
Konformne presl. (C-R), Lomljena Linearna presl. / Mobiusova presl.

Preslikava: $w = f(z) = u(x, y) + iv(x, y)$
 $f(z) = \frac{az + b}{cz + d}; ad - bc \neq 0$

Integrali
Pravila integriranja

Substitucija: $\int f(g(x))g'(x)dx = \int f(u)du, \begin{pmatrix} f+g \\ fg \end{pmatrix}' = \begin{pmatrix} f' \\ f'g \end{pmatrix} + \begin{pmatrix} g' \\ fg' \end{pmatrix}$

Po delih: $\int u\,dv = uv - \int v\,du$
 $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

Parcialni ulomki: $\frac{P(x)}{Q(x)} = \sum \frac{A_i}{(x-a_i)^{n_i}}$
 $(f(g(x)))' = f'(g(x)) \cdot g'(x)$

Elementarni integrali

$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$

$\int \frac{1}{x} dx = \ln|x| + C$

$\int e^x dx = e^x + C$

$\int \sin x dx = -\cos x + C$

$\int \cos x dx = \sin x + C$

$\int \frac{1}{1+x^2} dx = \arctan x + C$

$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$

Odvodi
Pravila odvajanja

Elementarni odvodi

$\frac{d}{dx} x^n = nx^{n-1}$

$\frac{d}{dx} e^x = e^x$

$\frac{d}{dx} \ln x = \frac{1}{x}$

$\frac{d}{dx} \sin x = \cos x$

$\frac{d}{dx} \cos x = -\sin x$

$\frac{d}{dx} \tan x = \sec^2 x$

$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$

$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$

Trigonometrske funkcije

$\sin^2 x + \cos^2 x = 1$

$\tan x = \frac{\sin x}{\cos x}$

$\sin 2x = 2 \sin x \cos x$

$\cos 2x = \cos^2 x - \sin^2 x$

$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$

$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$

$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$

Adicijski izrek za sin,cos :

$\sin(x + iy) = \sin x \cosh y + i \cos x \sinh y$

$\cos(x + iy) = \cos x \cosh y - i \sin x \sinh y$